Abiniya M

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OBJECTIVE

Passionate and detail-oriented Electronics and Communication Engineer with an interest in core electronics, focused on learning new competencies and expanding technical knowledge in a reputed organization.

EDUCATION

K.S.Rangasamy College of Technology
B E Electronics and Communication Engineering - CGPA: 8.75 (Upto 6th sem)

Kurinchi Matriculation Higher Secondary School
HSC — 93% (2022–2023) | SSLC — All Pass (2020–2021)

SKILLS

Technical Skills: Circuit simulation, Signal analysis using MATLAB, Basics of Machine Learning
Simulation Software: Multisim, MATLAB, Xilinx (basic exposure)
Software Skill: C programming

MINI PROJECTS

IoT-Based Smart Parking System Using ESP32 and Blynk July 2026
- Detects vehicles, checks availability of two parking slots, controls the gate automatically, and displays parking status on an LCD, with real-time monitoring via the Blynk app

AI Crop Recommendation System for Farmers Jan 2026
- Suggests the best crop using soil and weather data; improves yield through ML-based crop and fertilizer predictions

Heartbeat Sensor Using LM 358 Apr 2025
- Detects heartbeat signals from the fingertip; LM358 amplifies and filters the signal, indicated via LED blink

15W Audio Amplifier Using TDA2030 IC Apr 2025
- Implemented signal amplification, feedback network design, and speaker interfacing for high-quality, low-distortion audio output

ML Model for Heart Disease Prediction Using MATLAB Oct 2025
- Trained and evaluated multiple classification algorithms using the MATLAB Classification Learner App for accurate prediction

ACHIEVEMENTS

- **"An interpretable multi crop and fertilizer recommendation system using ensemble machine learning techniques"** accepted for publication, ICRETM 2026 — Current Science, 2026
- Won **1st prize** in MATLAB domain hackathon (skill-based training program)
- Participated in Hackforge 2.0 (Maatram Foundation NGO) — "AI Crop Recommendation for Farmers"
- Participated in MSME 4.0 hackathon — "Smart Surveillance and Security Solution for Fishery Ponds using IoT"
- Won **3rd prize** for paper "Hidden Camera Detector and Destroyer using LASER" at Kings College of Engineering
- Shortlisted in first 10 teams, Hackforge hackathon — "VR-Based Learning Tool for Children with ASD"

CERTIFICATION COURSES

- Employability Skills: Job Ready (with AI) — Wadhwani Foundation, July 2026
- Human-Computer Interaction (Top 5%) — NPTEL, May 2026
- Credit course on MATLAB with Machine Learning — Nov 2025
- Fundamentals of Machine Learning — Sept 2025
- Introduction to Industry 4.0 and IoT — NPTEL
- MATLAB Onramp — July 2025

EXTRACURRICULAR ACTIVITIES

- NSS Volunteer | IETE Club — Treasurer | Volunteer at Maatram Foundation (NGO)